

CHERVYANKA, Russian Federation

WMO: 293930

Lat: 57.654N

Lon: 99.542E

Elev: 724

StdP: 14.32

Time Zone: 8.00 (E08)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.6	-40.5	-47.8	0.3	-44.1	-43.4	0.5	-39.2	19.7	12.0	17.3	13.3	0.1	250	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	85.7	65.6	82.3	64.7	78.8	63.2	68.7	79.8	66.7	77.8	65.1	75.4	6.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	94.9	73.1	62.9	88.3	70.6	61.2	82.9	69.3	33.4	78.9	31.8	77.9	30.6	75.9	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 19.0	16.0	13.8	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-47.6	72.7	4.1	1.1	-50.6	73.4	-53.0	74.1	-55.3	74.7	-58.3	75.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.5	-8.6	-2.1	17.3	34.6	45.6	60.1	64.1	59.5	44.1	30.5	7.8	-1.3
	DBStd	27.73	17.65	14.72	13.42	8.39	8.09	7.48	5.12	6.04	7.71	9.73	16.07	17.65
	HDD50	8630	1817	1458	1014	465	191	8	0	6	210	607	1265	1589
	HDD65	13068	2282	1878	1479	912	605	182	78	186	626	1071	1715	2054
	CDD50	1143	0	0	0	4	53	311	438	302	34	1	0	0
	CDD65	104	0	0	0	0	2	35	50	17	0	0	0	0
	CDH74	1979	0	0	0	0	42	819	719	386	13	0	0	0
	CDH80	535	0	0	0	0	8	245	187	95	0	0	0	0
	WSAvg	5.1	3.9	3.6	5.1	7.3	7.6	5.4	4.4	4.1	5.0	5.7	5.0	4.2
	PrecAvg	12.2	0.5	0.4	0.3	0.6	1.1	1.3	2.2	2.3	1.3	0.8	0.8	0.7
(15) Precipitation	PrecMax	18.0	1.1	0.7	0.9	1.0	2.5	2.5	5.6	5.1	2.9	1.7	2.0	1.3
	PrecMin	7.8	0.2	0.0	0.0	0.0	0.3	0.3	0.4	0.3	0.4	0.1	0.3	0.1
	PrecStd	2.3	0.2	0.2	0.2	0.3	0.5	0.6	1.2	1.1	0.6	0.4	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.0	33.2	48.8	66.3	79.3	89.7	89.4	87.5	76.3	59.0	43.9	34.3
		MCWB	28.6	28.5	37.8	47.7	57.3	66.5	68.5	66.8	61.0	48.4	40.5	32.0
	2%	DB	26.4	29.5	43.4	60.7	72.9	86.2	85.0	82.5	69.3	52.8	35.5	29.1
		MCWB	24.7	26.9	34.8	44.7	53.4	65.0	66.5	64.9	56.7	43.7	32.8	27.0
	5%	DB	21.3	25.7	39.6	55.0	68.3	82.8	81.5	78.3	62.8	47.8	32.6	26.5
		MCWB	19.4	23.1	32.7	42.2	51.8	63.8	65.5	63.1	53.4	41.4	30.6	25.1
	10%	DB	17.0	21.1	36.4	50.2	63.1	79.0	78.0	74.5	57.6	43.3	29.3	23.2
		MCWB	15.6	18.8	30.9	40.4	49.0	62.5	64.2	62.0	50.8	38.4	27.2	21.5
	0.4%	WB	29.2	29.6	39.1	50.0	58.8	70.1	71.6	69.3	61.8	49.2	40.4	32.2
		MCDB	30.6	32.2	46.0	65.0	75.7	83.7	83.3	81.4	72.8	56.0	43.5	34.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	24.9	27.0	35.7	45.6	55.8	67.6	69.6	66.7	57.5	44.6	33.4	27.2
		MCDB	26.8	29.8	42.6	57.7	68.2	80.1	79.6	79.4	66.4	51.8	34.6	28.6
	5%	WB	19.7	23.1	33.2	43.1	53.0	65.7	67.5	64.8	54.8	41.5	30.4	25.1
		MCDB	21.4	25.7	39.0	54.2	65.0	78.5	77.1	75.4	60.8	46.8	32.3	26.6
	10%	WB	15.7	18.8	30.9	40.3	50.5	63.8	65.7	63.0	51.3	38.8	27.3	21.6
		MCDB	17.1	21.0	36.3	49.3	62.0	76.5	74.8	71.8	57.2	43.1	29.3	23.3
	5% DB	MDBR	19.4	25.8	25.8	20.2	22.6	26.7	23.3	24.0	19.8	15.5	16.9	17.3
		MCDBR	15.4	23.7	26.6	28.6	32.1	34.3	31.5	32.6	31.3	24.5	11.0	13.7
		MCWBR	14.0	20.7	19.3	16.0	16.3	15.6	14.7	16.8	18.6	16.3	10.4	12.3
		MCWBR	15.7	22.6	23.2	25.4	27.7	29.6	24.7	28.5	26.4	22.3	11.8	12.3
(35) Mean Daily Temperature Range	5% WB	MCWBR	14.3	19.9	17.6	15.8	15.0	14.2	12.9	15.5	16.5	15.8	11.0	11.6
	taub		0.284	0.311	0.340	0.374	0.400	0.436	0.453	0.420	0.357	0.340	0.300	0.270
		taud	1.993	2.011	2.019	2.151	2.241	2.210	2.184	2.242	2.393	2.260	2.059	1.960
(42) Clear-Sky Solar Irradiance	Ebn at Noon		166	220	249	263	264	255	247	247	248	213	158	130
		Edh at Noon	19	30	39	40	40	42	42	37	27	23	18	15
	RadAvg		215	521	974	1444	1630	1859	1722	1367	839	406	209	129
(44) All-Sky Solar Radiation	RadStd		25	37	57	85	124	145	113	101	115	37	28	21

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page